

Subgoal Scale and Reality-Verified Speculative Consumption in Hierarchical Latent World-Model Planning

Preliminary draft — PushT results only

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Status: preliminary draft, PushT-only. All PushT results below (Section 4–Section 11) are measured and cross-checked against raw logs. Section 13 (Reacher, TwoRoom, OGBench-Cube) is not yet run — placeholders only. Section 1 (contributions framing), Section 2 (related work), Section 3 (formal method description), and Section 14–Section 15 (discussion/conclusion) are lightly written and will be rewritten once the generalization results land; treat them as scaffolding, not final claims. One open data gap: the random-initialization table (Section 7) reports only the seeds we could directly verify against logs — see the note there.

Abstract

We study the FF-JEPA hierarchical latent planner — a frozen LeWM JEPA encoder, a diffusion subgoal drafter (“GDM”), and a CEM controller — on PushT, and identify *subgoal scale*, the spacing S between drafted subgoals with the control loop matched to it, as the dominant unexplored lever. Success rate over stride is non-monotone with a clear interior optimum at $S=10$: 97.2%/96.4% (20°/5° criteria) versus 91.1%/86.3% for the paper’s native $S=25$ under an identical protocol. On the paper’s own published protocols the winning configuration **exceeds the published numbers** at both horizons (96.1→98.4 at $t=25$, 91.8→97.9 at $t=75$, 20°, $n=256$); on random initialization (its hardest setting, published 82.4%) it records **zero failures in every verified episode measured so far** (512/512 across two seeds at $S=10$; a third arm and criterion cell are queued but unconfirmed, see Section 7); and on a fixed-population horizon sweep the baseline decays monotonically while $S=10$ stays flat — **the gap widens from +3.3 to +11.9pp**. A cadence-isolation control shows the mechanism is *scale matching*, not re-planning frequency. Separately, the one component of DSpark-style speculative decoding that survives measurement — a draft–verify–accept skeleton with the learned verifier replaced by the *achieved environment state* — cuts diffusion compute $\sim 11\times$ versus every-step drafting while remaining success-rate-neutral at the official criterion. Refinement heads, learned confidence scheduling, blind multi-step commitment, and a deterministic regressor that dominates the drafter *offline* each measurably *hurt* in closed loop; we report them as negative results that causally implicate drafter stochasticity.

1 Introduction

Hierarchical latent world models promise long-horizon planning by decomposing a goal into a chain of reachable subgoals. FF-JEPA [TBD, TBDc] instantiates this on top of the LeWM JEPA world model [TBD, TBDd]: a diffusion model drafts future subgoal latents and a flat CEM controller [TBD, TBDa] steers toward each in turn, re-planning on a fixed stride. It reports that hierarchy beats flat CEM on long horizons — but under an underspecified pipeline (no released code; unstated diffusion configuration; unstated evaluation population), and with a subgoal stride fixed at a single value.

We take FF-JEPA as a substrate and ask what actually governs its closed-loop success. Our contributions:

1. **Faithful reproduction and audit** (Section 4). We rebuild the pipeline to the episode and iteration, reproduce the paper’s headline numbers, and establish that the benchmark’s *coded* success criterion is 20° angular tolerance (its 5° appears in prose only); we report both throughout.
2. **The subgoal-scale law** (Section 5). Retraining the drafter at $S \in \{5, 10, 15, 25\}$ with the control loop matched to the stride reveals an interior optimum at $S=10$ that beats the paper configuration by $+6.1 / +10.1$ pp and, on the paper’s own protocols, exceeds its published numbers. A cadence-isolation control identifies *scale matching*, not re-planning frequency, as the mechanism.
3. **Reality-verified speculative consumption** (Section 8). Replacing DSpark’s learned verifier with the achieved environment state yields an $\sim 11\times$ reduction in diffusion compute at neutral success rate.
4. **Negative results** (Section 6, Section 9). Refinement heads, blind commitment, learned confidence, and a deterministic regressor all hurt in closed loop despite matched-or-better *offline* fidelity — causally implicating the drafter’s conditional *spread* as load-bearing.

2 Related Work

[TBD — Related work. Placeholder pointers only — full positioning pending a literature-occupancy pass (flagged as an open thread, see Section 14). Axes to cover: (i) hierarchical / subgoal-conditioned latent planning (FF-JEPA [TBD, TBDc], LeWM [TBD, TBDd], and other JEPA-style world models); (ii) re-planning cadence and receding-horizon MPC in learned-model control, including CEM-based planners [TBD, TBDa]; (iii) speculative decoding, both in language models (the DSpark composition we port, [TBD, TBDb]) and prior applications of draft–verify–accept outside text generation; (iv) subgoal/waypoint spacing and hierarchical RL more broadly; (v) budget-controlled evaluation protocols for long-horizon manipulation, contrasting our fixed-population sweep with VLWM’s fixed-budget-50 regime [TBD, TBDf] and OGBench’s benchmark suite [TBD, TBDg].]

3 Method

We modify only three things about the FF-JEPA system (Section 4): the subgoal spacing the drafter is trained and queried at, the loop geometry that consumes its output, and the policy that decides when to re-invoke it. Encoder, drafter architecture, training objective, and CEM itself are untouched.

Scale-matched retraining. We retrain the GDM drafter at stride $S \in \{5, 10, 15, 25\}$ with an identical recipe to the paper’s $S=25$ configuration (loss, batch size, learning rate, schedule, epochs — Section 4); only the subgoal spacing used to build training pairs changes, so pair counts are construction-identical across arms and any SR delta is attributable to spacing alone. Because the CEM terminal cost is only well-posed when its rollout length is comparable to the target’s distance, the control loop is scaled with the stride: CEM horizon = replan cadence = S (holding the action-repeat block, sample count, and top- k fixed). This coupling is definitional, not a second free variable — see the cadence-isolation control in Section 5, which holds targets fixed at 25-step spacing while re-anchoring every 5 steps, and isolates loop cadence from target spacing as separate mechanisms.

Reality-verified speculative consumption. Section 8 details the mechanism; briefly, the drafter proposes an N -block once, the policy executes toward the first position, and after each hop compares the *achieved*

latent to the subgoal just pursued: within tolerance τ , the next drafted position is consumed with no diffusion call; otherwise the planner re-drafts from the achieved state. This differs from DSpark’s original scheme only in the verifier: DSpark scores acceptance with a learned confidence head trained on drafter self-consistency; we use the environment’s realized state directly, which is unbiased by construction and requires no additional training.

[TBD — Formal algorithm box / pseudocode for both procedures, and the CEM cost functional definition, are pending — currently described in prose above and in Sections 5 and 8.]

4 Setting and Faithful Reproduction

The FF-JEPA planner. Observations are encoded by a frozen LeWM JEPA encoder, $z_t = \text{enc}(o_t)$. A diffusion model G (“GDM”: a 3-block DiT with adaLN-Zero conditioning, $N=3$ future subgoals, ϵ -prediction, trained on success-filtered expert demonstrations) proposes the next subgoal latent $\hat{z}_{sg,m+1}$ from the current encoded observation. CEM then optimizes an action sequence against a terminal- L_2 cost to that subgoal, re-planning every H environment steps. The paper fixes the stride to $H=25$.

Reproduction (audit closed). Our faithful retraining reproduces the paper: short-horizon 95.31% (paper 96.09%; same-set oracle 93.36%; $n=256$), long-horizon $t=75$ multiseed mean 90.0% (paper 91.80%), random-init 96.5%/78.1% at $20^\circ/5^\circ$ (paper 82.42%). Pipeline faithfulness is over-determined: a from-scratch rebuild of the dense training set reproduces the original training population to the episode (9,094) and iteration (171,940), and the retrained $S=25$ drafter reproduces the reference offline fidelity anchor to four decimals (0.1593 vs. 0.1593 at $m+1$).

The $20^\circ/5^\circ$ criterion. FF-JEPA’s prose states a 5° angular tolerance, but the environment’s shipped evaluation code enforces 20° ; the oracle (true goal, exact pipeline) reaches only $\sim 84\text{--}86\%$ at literal 5° even under matched-population filtering, making a genuine 96% at 5° for a *sampled*-subgoal method implausible. We therefore treat 20° as the operative criterion and 5° as a strict *precision* criterion, reporting both throughout.

Protocols. Two protocols recur below. *Paper protocol*: $n=256$, paper-matched sampling (reproduction and random-init). *Controlled protocol*: $n=62$ fixed held-out episodes, goal offset 150, budget 300 steps, block-only criterion, CEM seeds 42–45 (curve: 42–49); every arm sees byte-identical episodes and seeds, so within-table deltas are exact treatment effects. Per-arm SE $\approx 1.7\text{pp}$ (4 seeds) / 0.9pp (8 seeds).

5 The Subgoal-Scale Law

We retrain the drafter at $S \in \{5, 10, 15, 25\}$ with an identical recipe — only the target spacing changes, and pair counts are construction-identical — and scale the control loop with the stride (CEM horizon = re-plan cadence = S). Offline pre-gates confirmed fine-stride targets are well-posed (displacement $4.7\times$ the encoder noise floor at $S=5$) and that the fine-scale CEM cost landscape *sharpens* (true-plan ranking percentile 0.93 at $S=5$ vs. 0.82 at $S=25$; cost dynamic range $3\times$ larger).

Mechanism: scale matching, not cadence. A cadence-isolation control — re-anchor every 5 steps against *unchanged* 25-step targets — *collapses* to 78.6%/77.8% (-12.5pp below baseline): each re-draft re-samples a distant subgoal from a wide conditional posterior, and CEM chases a churning target. Together with the

Table 1: Success rate vs. subgoal scale (controlled protocol). A clear interior optimum at $S=10$: +6.1/ +10.1pp over the paper configuration (+8.7pp at 5° with 8 seeds, $SE \approx 0.9$ pp, positive in every seed). $S=10$ pays almost no penalty between the loose and strict criteria; $S=25$ drops ~ 5 pp.

	$S=5$	$S=10$	$S=15$	$S=25$ (paper config)
SR 20° (seeds 42–45)	92.3	97.2	96.8	91.1
SR 5° (seeds 42–45)	91.9	96.4	94.4	86.3
SR 5° (8 seeds, 42–49)	92.9	96.4	94.4	87.7
Diffusion calls / episode	60	30	20	12

monotone commitment losses of Section 6, the correct statement is: **closed-loop success is governed by matching the control loop to the subgoal scale; finer matched scales win, to an interior optimum** — not “re-plan more,” not “re-plan less.”

Prediction error is horizon-intrinsic. Across all four drafters, offline prediction error collapses onto the *absolute* look-ahead horizon, independent of block position (+10 steps: 0.105 vs. 0.112; +15: 0.132 vs. 0.140), and the $S=5$ drafter’s *third* block position (+15, 0.140) outpredicts the $S=25$ drafter’s *first* (+25, 0.159). “Suffix decay” is task-future uncertainty, not an architectural defect — retiring intra-block architecture work.

6 Negative Results: The DSpark Port

We implemented the published DSpark [TBD, TBDb] speculative-decoding composition on the GDM and evaluated each component in the closed loop (controlled protocol, $S=25$).

Table 2: DSpark components ported literally. Lower latent error does not mean better control: the refiner *halved* offline prediction error while losing 11pp SR — offline fidelity anti-correlates with closed-loop SR. The mechanism is conditional-spread collapse: pulling drafts toward the conditional mean produces less-reachable targets. This closes refinement, learned-confidence scheduling, and blind commitment as directions.

Arm	SR 20°	SR 5°	Verdict
GDM, re-draft every step (reference)	91.9	83.1	—
+ refinement head (fixed-1, same cadence)	81.1	72.2	refinement –11pp
+ blind commit-3	73.8	50.0	commitment –18pp
+ learned-confidence adaptive commit	64.5	52.0	worse than blind
raw open-loop chain (commit-6)	72.6	63.7	healthy chain still loses

7 Random Initialization: The Paper’s Hardest Setting

Data-completeness note. A second $S=5$ seed and a third/fourth $S=10$ seed were queued (Isambard job 5578762) to extend this table to $n=256 \times 2$ ($S=5$) and $n=256 \times 4$ ($S=10$) — matching the power of the rest of the paper’s tables. We could not locate a result for that job in any local log, results file, or the remote cluster (unreachable at the time of writing) and do not report it. *Every number in this table is a directly*

Table 3: Random initialization, per-episode random goals (paper III-A protocol). The protocol bug that once produced a spurious 100% was fixed *before* these runs; the reproduced baseline scoring 78.1% on the identical code path is the control. $S=10$: 512/512 **episodes over two independent seeds, zero failures at both tolerances**. The paper’s criterion mapping is ambiguous for this cell (its 82.42 aligns with our strict-criterion baseline); under either reading, fine scale moves it to 100% on every seed measured so far.

	SR 20°	SR 5°
LeWM (flat, goal image; paper)	0.00	
FF-JEPA Det (paper)	81.25	
FF-JEPA GDM (paper)	82.42	
FF-JEPA GDM (our faithful repro, $n=256$, seed 42)	96.5	78.1
Ours, $S=5$ ($n=256$, seed 42)	100.0	94.5
Ours, $S=10$ ($n=256 \times 2$ seeds, 42–43)	100.0	100.0

verified success rate; do not cite figures beyond what is tabulated here (in particular, not “1024 episodes” or “four seeds”) until the queued arm is confirmed and this note is removed.

8 Reality-Verified Speculative Consumption

The surviving DSpark idea is its *skeleton*: draft a block cheaply, verify, accept the surviving prefix. Our port replaces the learned verifier with reality: draft the N -block once; after each hop, compare the *achieved* latent to the subgoal just pursued; within tolerance $\tau \Rightarrow$ consume the next drafted position (no diffusion call); otherwise $\Rightarrow$ re-draft from the achieved state. Re-anchoring is never sacrificed when reality diverges. Offline calibration showed 8-step DDIM drafts are indistinguishable from 50-step drafts (matched-noise distance 0.03; spread preserved); both transferred to the closed loop.

9 Is the Diffusion Drafter Necessary?

A deterministic MLP regressor ($z_{cond} \rightarrow$ block, the offline-strong JointMLP class, weight decay 10^{-3}) as the subgoal source, re-anchoring every replan, matches the diffusion drafter’s *offline* prediction error on the same windows (0.30 vs. 0.32 at $S=10$). Closed loop, it loses everywhere.

10 Failure Anatomy

Instrumented traces (every re-plan’s achieved latent and drafted subgoal) over all 8 seeds of $S=10$: **18 failures in 496 episodes, all tolerance-boundary near-misses** — final latent distance-to-goal within the success distribution’s 95th percentile (one failure sits *closer* than the median success: a pure angle miss). Zero unreachable-subgoal failures, zero divergence, zero stuck episodes; successful episodes’ commanded-to-achieved step ratio has median 1.06. FF-JEPA’s own reported failure mode (“subgoal looks right, agent cannot reach it”) is *eliminated* at fine scale; the residue is quantization at the success threshold.

11 Horizon Scaling (Fixed Population)

Goal offsets $t \in \{25, 50, 75, 100\}$ on the *identical* 256 episodes at every offset (reindexed from the $t=150$ set), budget $2t$, seeds 42/43 — so horizon is the only variable, free of the eligibility confound that invalidates

Table 4: Speculative consumption (controlled protocol). NFE = denoising function evaluations per episode (calls $\times$ DDIM steps). At 8-seed power, verified coasting is 20°-neutral at $S=10$ but costs $\sim 3.5\text{pp}$ at 5°; at $S=5$ the sign *reverses* (spec-accept beats its own every-step reference on both criteria) — when re-drafts churn a close target, skipping verified re-drafts is better control. The full $S=10$ system stays $+6.3/ +5.2\text{pp}$ ahead of the paper configuration (8 seeds) at $4.4\times$ lower diffusion cost.

Arm ($S=10$ unless noted)	SR 20°	SR 5°	call ratio	NFE/ep
every-step drafting, $k=50$ (reference)	97.2	96.4	1.00	1500
spec-accept $\tau=0.15$, $k=8$	98.0	95.2	0.67	161
spec-accept $\tau=0.20$, $k=8$	98.0	94.8	0.56	135
spec-accept $\tau=0.30$, $k=8$	96.8	94.8	0.45	108
spec-accept $\tau=0.20$, $k=50$ (k-isolation)	98.0	95.6	0.55	825
blind commit-2 (no verification)	98.8	96.4	0.50	750
$S=5$ every-step (reference)	92.3	91.9	1.00	3000
$S=5$ spec-accept $\tau=0.20$, $k=8$	98.0	94.8	0.42	202
<i>8-seed extension (seeds 42–49; SE $\approx 0.9\text{pp}$)</i>				
$S=10$ every-step	98.0	96.4	1.00	1500
$S=10$ spec-accept $\tau=0.20$, $k=8$	97.4	92.9	0.56	135
$S=5$ every-step	92.9	92.9	1.00	3000
$S=5$ spec-accept $\tau=0.20$, $k=8$	98.4	96.2	0.42	202
paper config ($S=25$ every-step, 8 seeds 5°)	91.1	87.7	—	600

independent per-offset draws.

Cross-paper regime (VLWM fixed-budget-50). We additionally evaluate under the VLWM [TBD, TBdf] fixed-budget-50 regime — a harsher protocol than FF-JEPA’s final-window budget of $2t$, holding the step budget at 50 for *every* offset (Table 7).

Four read-offs, all consistent with the main story: (a) **budget starvation reproduces the collapse** — confirming VLWM’s PushT cliff is largely a budget artifact, which validates our protocol note; (b) **within their regime we sit $\sim 3\times$ above their bars** (60.9 vs. 20 at $t=75$), citable as “same budget regime” with the start/goal-sampling caveat; (c) **spec-accept beats every-step at every offset** under time pressure — the anti-churn effect (Section 8) generalizes to $S=10$ once budget binds, a new data point for the τ -trade; (d) **the $S=10$ – $S=25$ gap compresses under starvation** — scale matching buys precision, not raw speed.

12 Headline Comparison

13 Generalization to Further Environments

Every result above is on PushT. To establish that the subgoal-scale law and reality-verified speculative consumption are properties of hierarchical latent planning rather than of one task, we extend to three environments of increasing structural distance from PushT. *These runs are in progress; tables below are placeholders.*

Table 5: Diffusion vs. deterministic regression as the subgoal source (seeds 42–45). **The drafter’s stochasticity is load-bearing:** at matched-or-better offline fidelity, the zero-spread mean target loses everywhere, and its -11.7pp strict-criterion loss at $S=25$ quantitatively mirrors the refinement head’s -11pp (Table 2), closing the spread thesis causally — any operator that regresses subgoals toward the conditional mean damages control.

	$S=10$		$S=25$	
	20°	5°	20°	5°
diffusion drafter (reference)	97.2	96.4	91.1	86.3
deterministic regressor	91.9	91.9	86.3	74.6
Δ	-5.2	-4.4	-4.8	-11.7

Table 6: Fixed-population horizon sweep ($n=512/\text{cell}$). **The baseline decays monotonically at the strict criterion (-11.9pp from $t=25$ to $t=100$) while $S=10$ is flat from $t=50$ onward — the gap widens monotonically.** Spec-accept tracks every-step within 1.2pp across the curve (call ratio ≈ 0.57 at long horizons, ≈ 0.85 at $t=25$: the cost win is a long-horizon phenomenon).

	$S=25$ (FF-JEPA)		$S=10$ every-step		$S=10$ + spec-accept	
t	20°	5°	20°	5°	20°	5°
25	98.1	93.8	99.8	97.1	99.6	95.1
50	93.6	85.0	97.5	93.8	97.1	93.9
75	92.6	83.2	98.2	93.8	97.9	93.0
100	93.8	81.8	97.9	93.8	98.6	92.6
5° gap ($S=10$ – baseline)			$+3.3 \rightarrow +8.8 \rightarrow +10.5 \rightarrow +11.9$			

13.1 Reacher

A low-dimensional continuous-control reach task — the recon that decides whether the scale law transfers to a non-pushing morphology. Same protocol: retrain the drafter at $S \in \{5, 10, 15, 25\}$, report SR vs. scale and the spec-accept efficiency arm.

[TBD — Reacher: SR vs. subgoal scale + spec-accept NFE. Recon in progress.]

13.2 TwoRoom

A navigation task with a bottleneck between rooms, probing whether scale matching holds when the optimal subgoal density is spatially non-uniform.

[TBD — TwoRoom: SR vs. subgoal scale under goal-conditioned planning.]

13.3 OGBench-Cube

A manipulation benchmark [TBD, TBDe] for the strongest generalization claim: multi-object state and longer effective horizons, where the $\sim 11\times$ diffusion-cost reduction matters most.

[TBD — OGBench-Cube: headline SR / NFE vs. FF-JEPA baseline.]

Table 7: Cross-paper cell: our three arms under VLWM’s fixed-budget-50 regime, all 20° , seed means. Rightmost columns are the published PushT numbers of VLWM and of LeWM (flat) under that regime. Budget starvation reproduces the published collapse, yet within the identical budget our arms sit $\sim 3\times$ above VLWM’s bars. Curve *shapes* are comparable across papers; absolute numbers are not (start/goal sampling differs), so we cite this as “same budget regime,” not a like-for-like win. [†]At $t=25$ the fixed-50 budget equals FF-JEPA’s native $2t$ budget, so the two protocols coincide (the row is the $t=25$ cell of Table 6); the budget binds only for $t \geq 50$, which is where the published VLWM/LeWM bars begin.

t	Ours (fixed-budget-50)			Published	
	$S=25$	$S=10$	$S=10 + \text{spec}$	VLWM	LeWM (flat)
25 [†]	98.1	99.8	99.6	94	90
50	83.6	88.7	90.0	60	46
75	55.7	59.8	60.9	20	12
100	23.4	24.4	27.3	12	8

14 Discussion and Limitations

What the story is. Closed-loop success in this planner is governed by matching the control loop to the subgoal scale, and the drafter’s conditional *spread* is load-bearing: every operator we tried that collapses drafts toward the conditional mean (refinement, blind commitment, deterministic regression) damaged control despite equal or better offline fidelity. The efficiency win is orthogonal — reality is a free verifier.

Limitations. (i) FF-JEPA is a preliminary paper with no released code; our “faithful” configuration is a principled reconstruction where the paper is silent, not a verified match on every axis. (ii) All confirmed results are on PushT; cross-environment generality (Section 13) is **TBD**— Reacher and TwoRoom drivers are implemented and committed but not yet run (pending cluster availability); OGBench-Cube is not yet started. (iii) The $20^\circ/5^\circ$ criterion mapping for the paper’s own cells is inferred from empirical alignment, not stated by the paper; we mark every cross-paper comparison with the side it is valid on. (iv) The $t=150$ cell is a beyond-paper probe, not a reproduction claim. (v) The random-initialization table (Section 7) is one arm short of its intended power — a queued confirmation run ($S=5$ second seed, $S=10$ third/fourth seeds) has no recoverable result and is reported as such rather than assumed.

15 Conclusion

[TBD — This is a placeholder conclusion for a preliminary, PushT-only draft — it will be rewritten once Section 13 has real numbers and is not meant to stand as this paper’s final framing.]

On PushT, subgoal scale matched to the control loop is the strongest lever we found in FF-JEPA-style hierarchical latent planning: a fine matched stride ($S=10$) beats the paper configuration across every setting we measured and, on the paper’s own protocols, exceeds its published numbers. Reality-verified speculative consumption recovers $\sim 11\times$ of the added diffusion cost at neutral success rate. Whether either claim is a property of hierarchical latent planning in general, versus an artifact of PushT specifically, is exactly what Section 13 is for and is not yet answered — that is the remaining work, not a formality.

Table 8: Where we stand, both criteria per cell (20° = the criterion enforced by the benchmark’s shipped code; 5° = the strict precision criterion its prose states). “Ours” modifies FF-JEPA’s GDM system only in subgoal scale, loop matching, and consumption policy — encoder, drafter architecture, training recipe, and CEM are unchanged. ^pThe paper publishes one number per cell without stating its criterion; each is placed on the side it empirically aligns with, with $\cdot$ marking the untested side. Comparisons are valid only within a side.

Setting (SR 20° / 5°)	LeWM (flat)	FF-JEPA GDM		Ours	
		published ^p	our repro	$S=10$	+ spec-accept
Short horizon, $t=25$	94.5	96.1 / $\cdot$	95.3 / $\cdot$	98.4 / 94.1	98.4 / 95.3
Long horizon, $t=75$	3.5	91.8 / $\cdot$	90.0 / 78.8	97.1 / 93.8	97.9 / 91.6
Long horizon, $t=150^a$	—	—	91.1 / 87.7	98.0 / 96.4	97.4 / 92.9
Horizon sweep, $t=100^b$	—	—	93.8 / 81.8	97.9 / 93.8	98.6 / 92.6
Random initialization ^c	0.0	$\cdot$ / 82.4	96.5 / 78.1	100.0 / 100.0 ^d	—
Diffusion NFE / episode ^a	0	—	600	1500	135

^aControlled protocol, 8 seeds. ^bFixed-population sweep, $n=512$ /cell. ^c $n=512$ over 2 seeds, zero failures at both tolerances; spec-accept not run in this mode (the drafter is the tested component). ^dTwo further seeds are queued but unconfirmed — see the data-completeness note in Section 7; do not extrapolate beyond the verified $n=512$.

References

- TBD. Cross-entropy method for planning/control (placeholder citation). *TBD*, TBDa.
- TBD. DSpark speculative decoding (placeholder citation – fill with the real title/authors/venue). *TBD*, TBDb.
- TBD. FF-JEPA (placeholder citation – fill with the real title/authors/venue). *TBD*, TBDc.
- TBD. LeWM (placeholder citation – fill with the real title/authors/venue). *TBD*, TBDd.
- TBD. OGBench (placeholder citation – fill with the real title/authors/venue). *TBD*, TBDc.
- TBD. VLWM (placeholder citation – fill with the real title/authors/venue). *TBD*, TBDf.